

Final paper

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Introduction

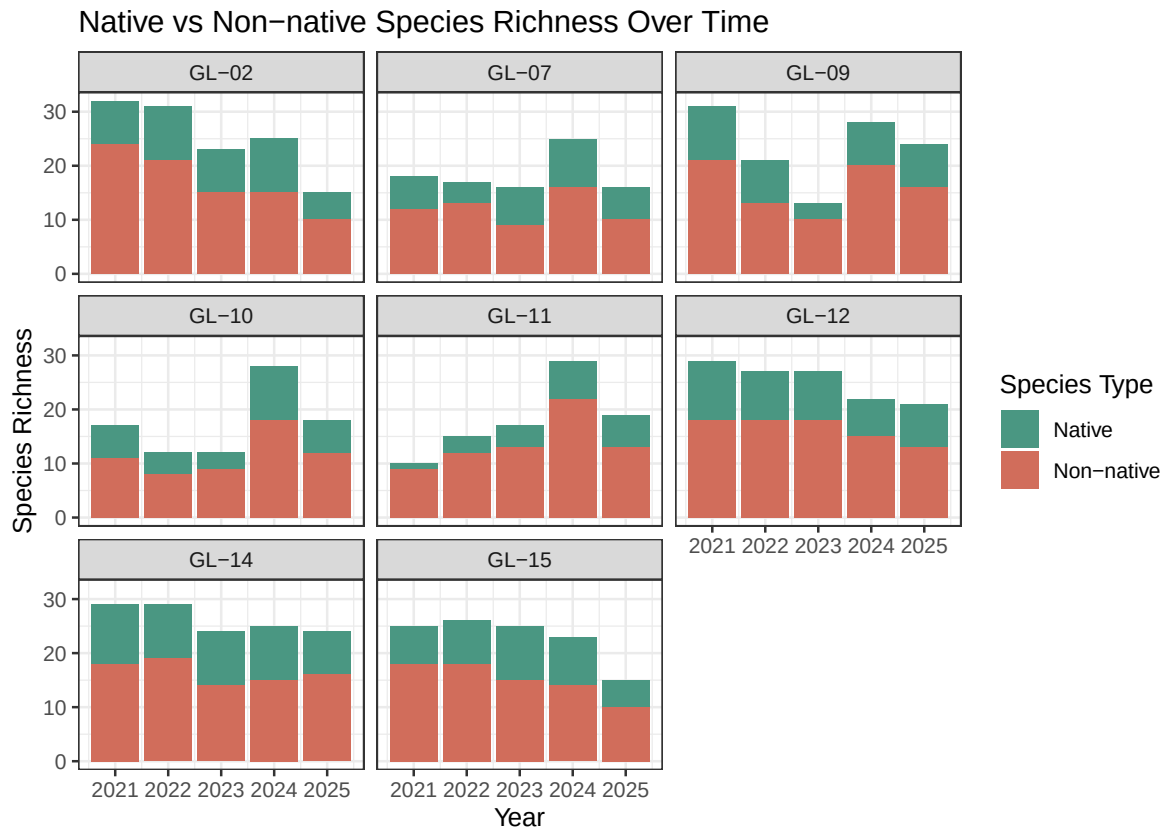
Methods

To answer our research question, we used a vegetation dataset sourced from the Cheadle Center for Biodiversity and Ecological Restoration (CCBER), and provided to us by the ENVS 193DD course. This dataset spans 5 years (2021-2025) of vegetation data from across several habitats in North Campus Open Space (NCOS) including grasslands, mosaic habitats, wetlands, open water, woodlands, and shrublands.

We chose to focus on native perennial grasslands, which encompasses measurements from 8 different transects across the 16.8-acre mesa. The method of data collection used for vegetation surveys is the Quadrat Transects (QT) method. This method involves the use of 11 one-square-meter quadrats placed by a 30-meter transect, with quadrats alternating between the left and right side of the transect line every 3 meters. “The quadrats are subdivided into 100 ten-centimeter squares and the percent cover is estimated [by human monitors] for each species in the quadrat,” (Rickard 2024).

We will not be performing a formal statistical analysis for our project because we are primarily exploring a correlative relationship rather than predicting trends for species cover and richness. However, we will be identifying major events that might influence a directional response in vegetation percent cover and species richness, and examining whether a response was captured in our dataset. Major events include the prescribed burn in Fall 2023 because of its effects as an intermediate disturbance event, as well as wet (2023, 2024) and dry (2021, 2022) water years because water is a limiting resource for key native grass species (*S. pulchra*). These factors are potential influences on how percent cover and species richness change over time.

Results



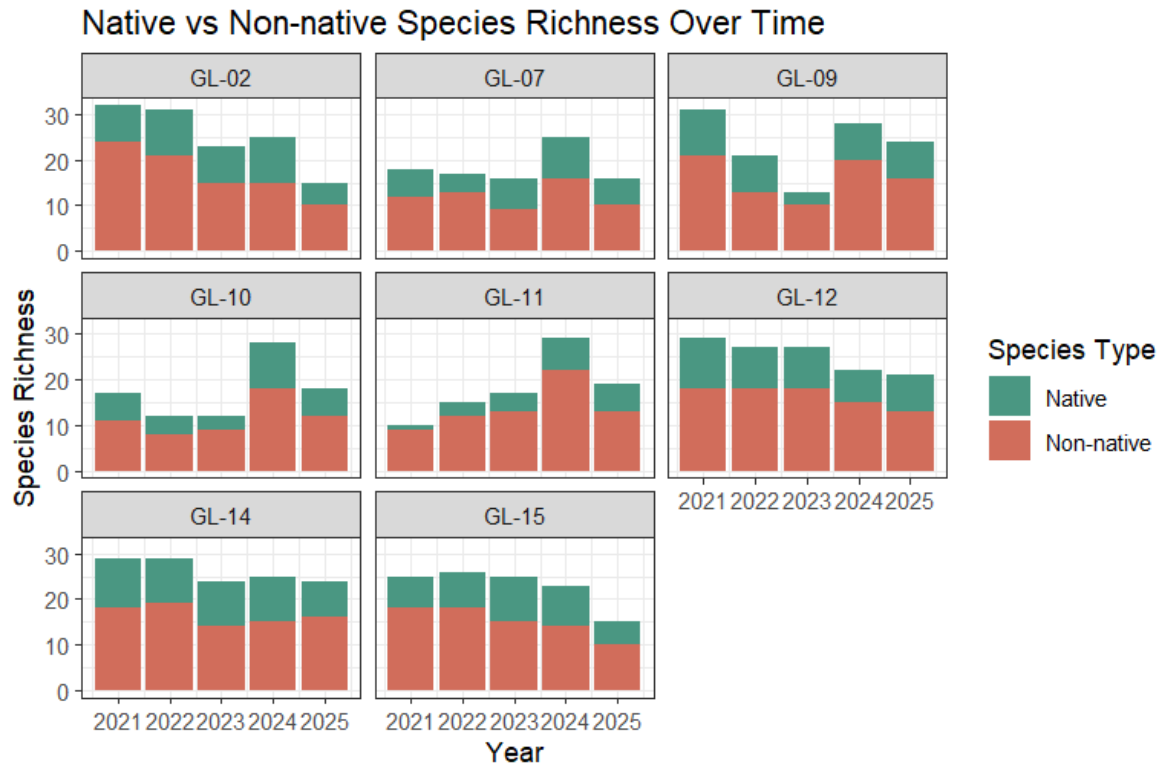
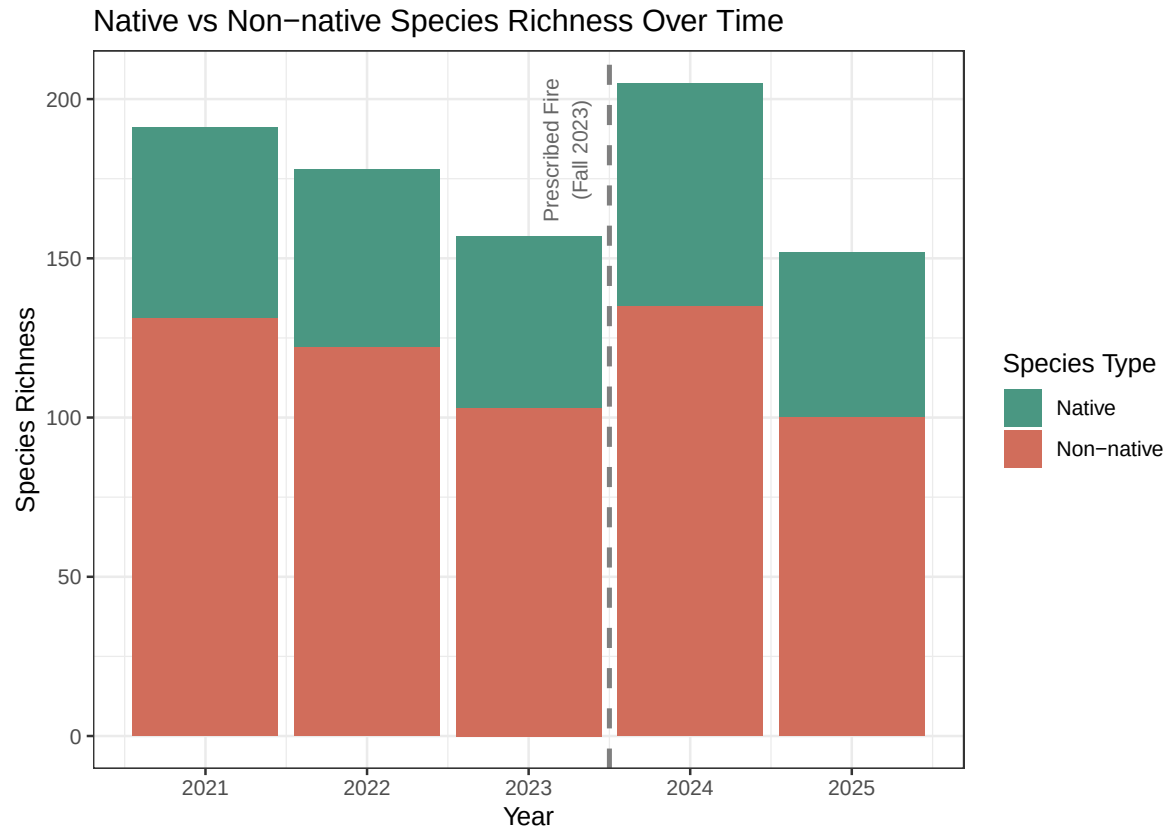


Figure 1: Figure 1



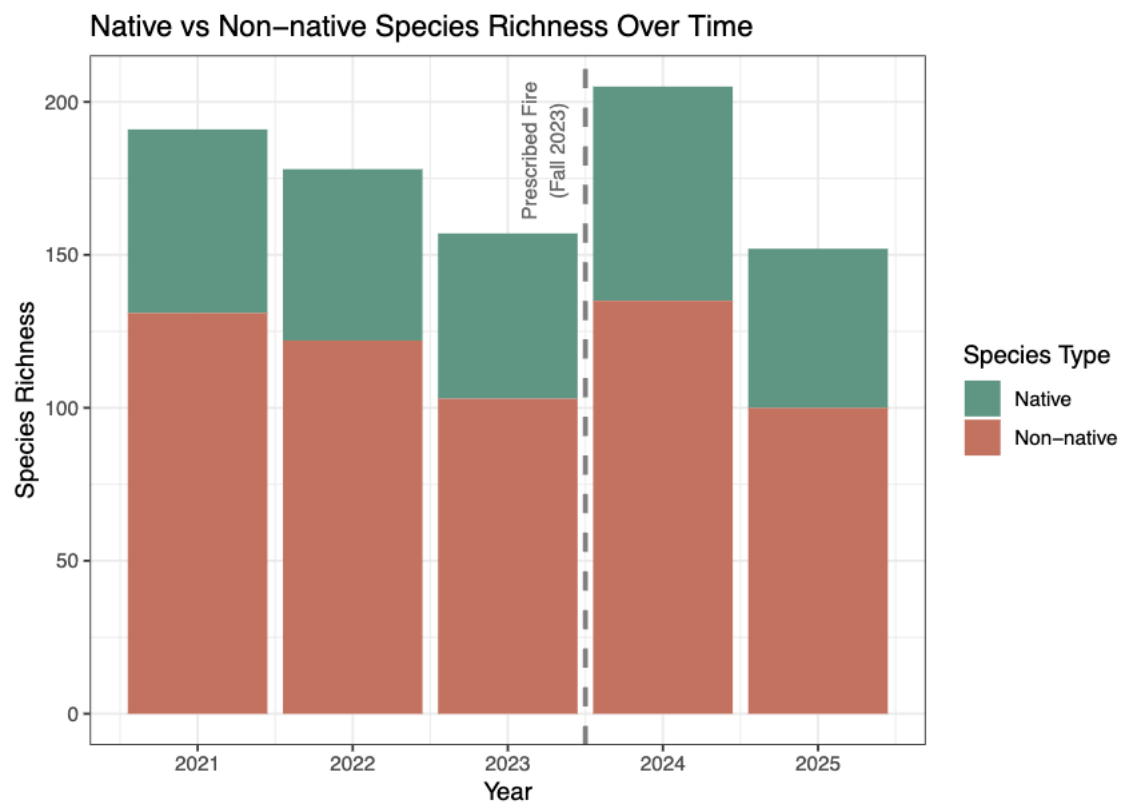
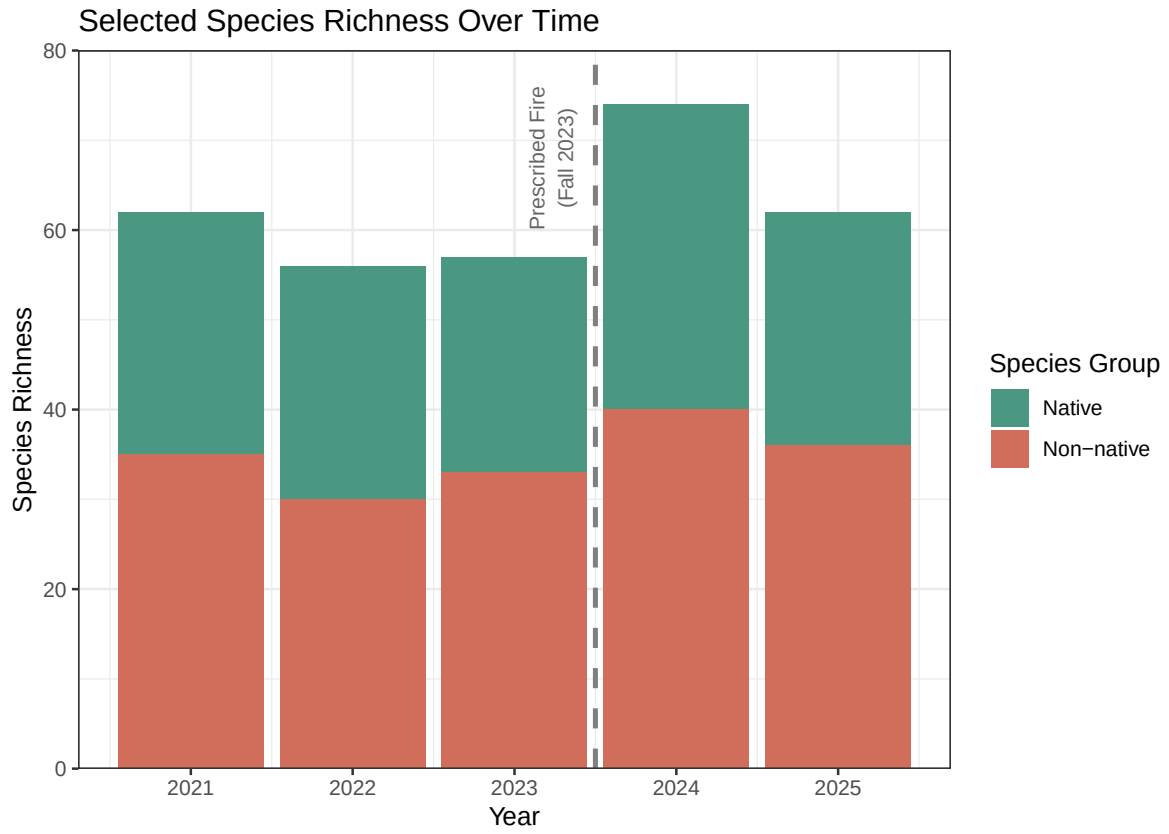


Figure 2: Figure 2



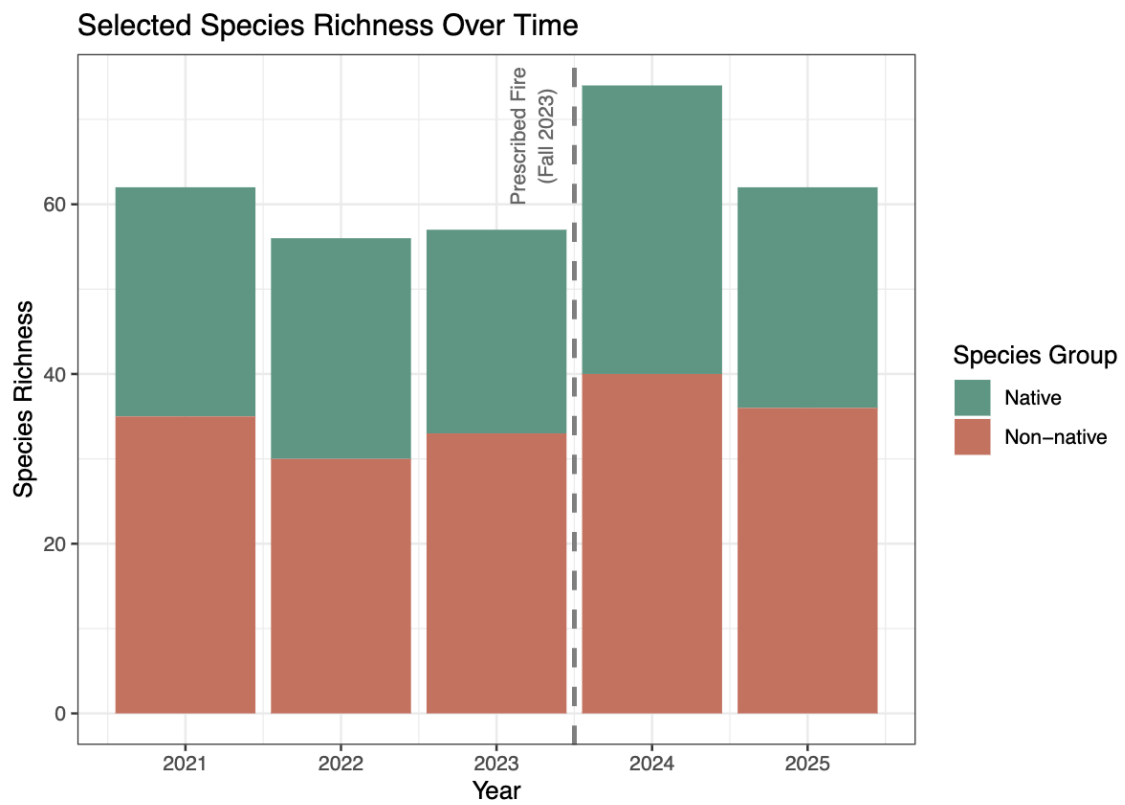
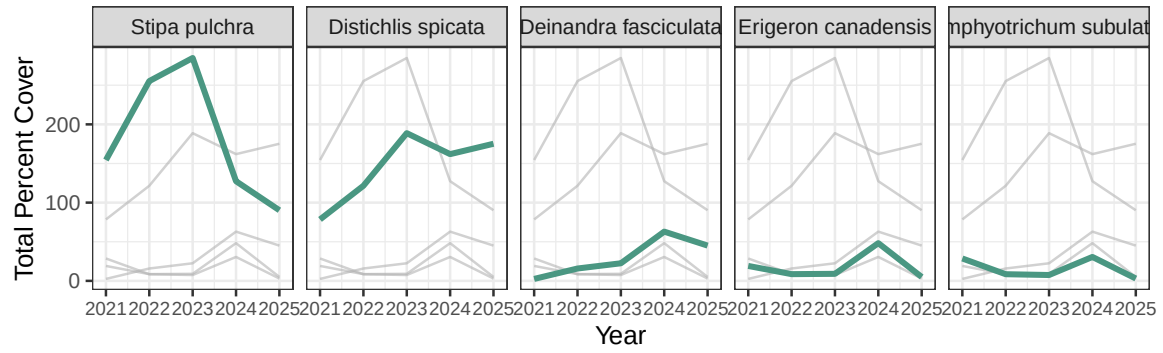
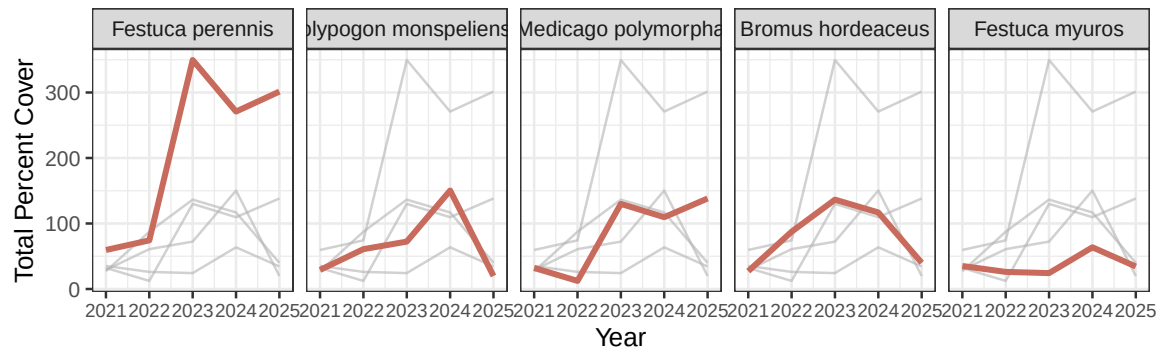


Figure 3: Figure 3

Selected Native Species Percent Cover Over Time



Selected Non-native Species Percent Cover Over Time



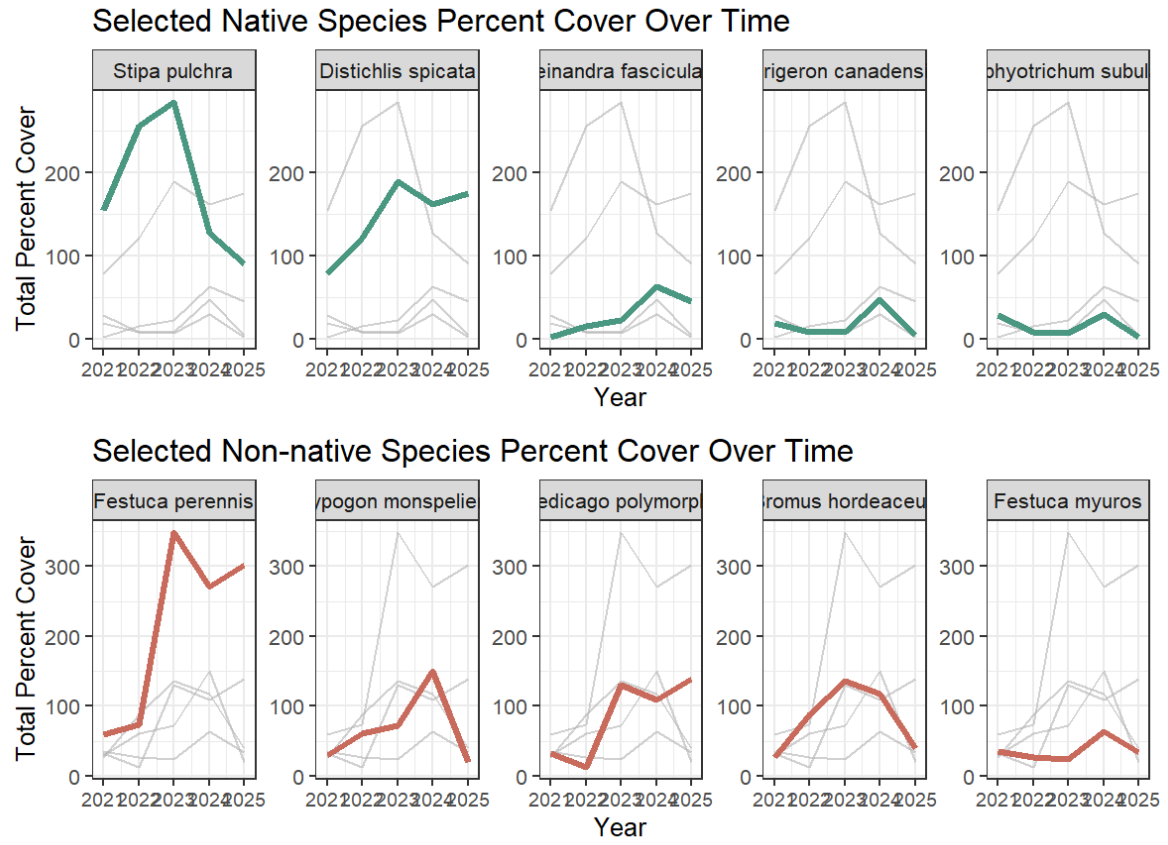
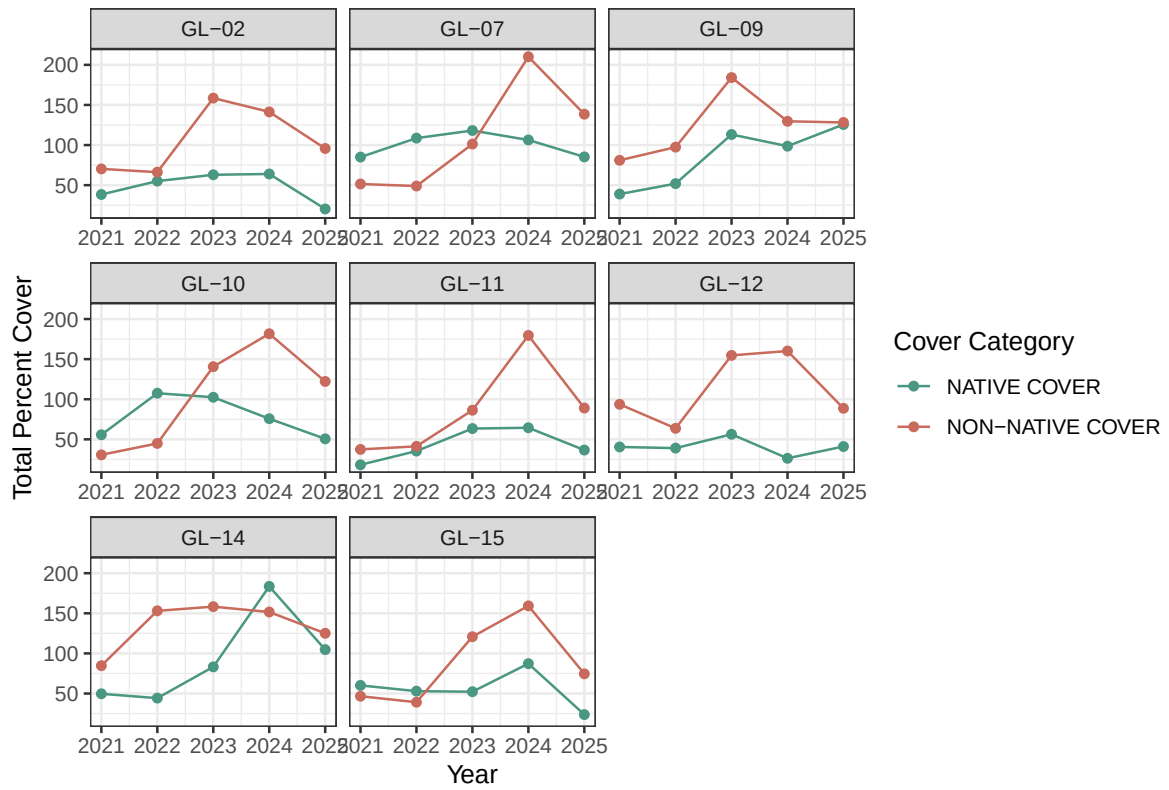


Figure 4: Figure 4

Native vs Non-Native Percent Cover Over Time Across GL Transects



Native vs Non-Native Percent Cover Over Time Across GL Transects

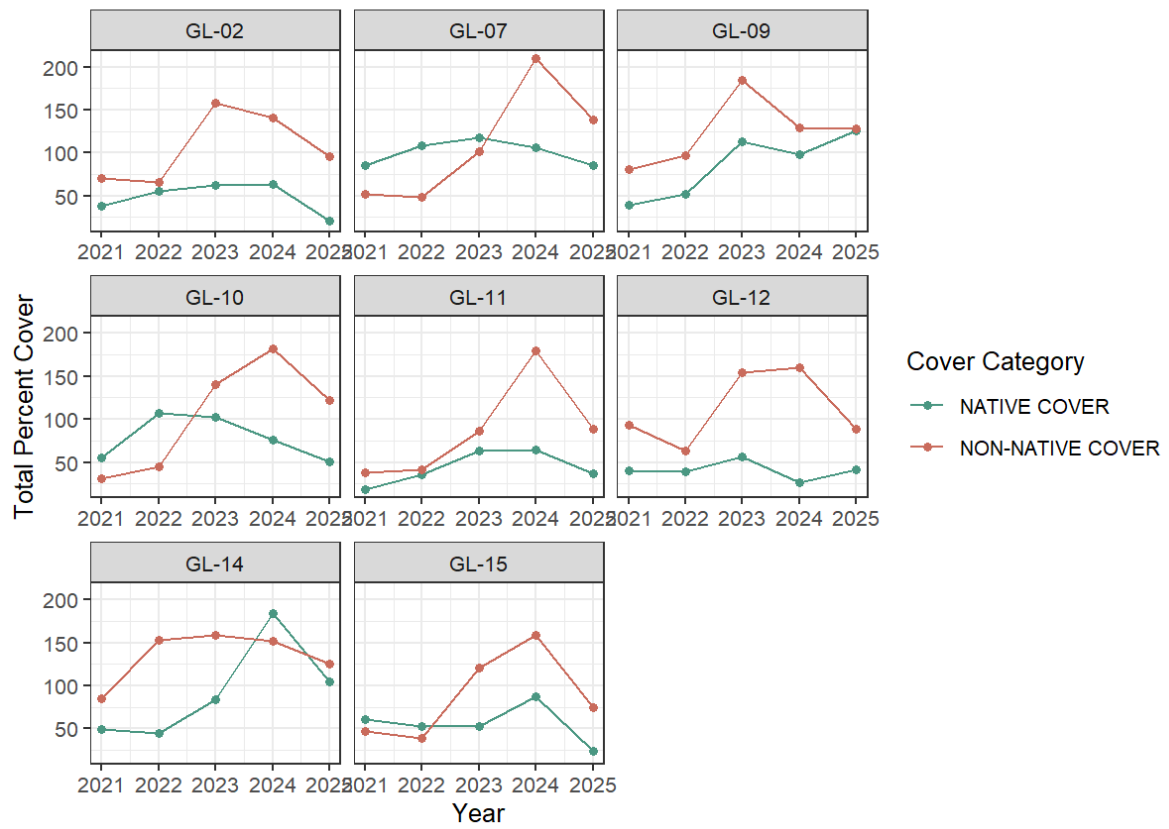
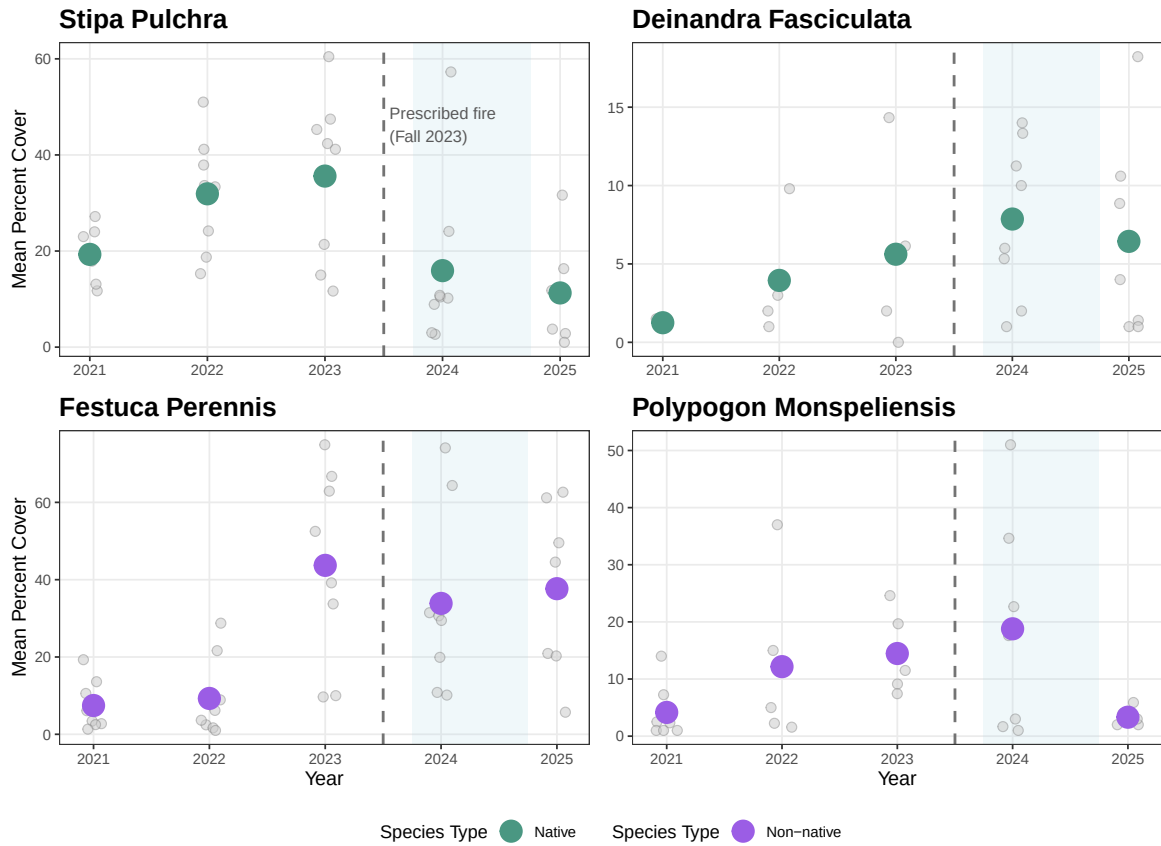


Figure 5: Figure 5

Percent Cover of Four Keys Species Through Time

Blue shading indicates the 2024 water year (143% of normal precipitation).



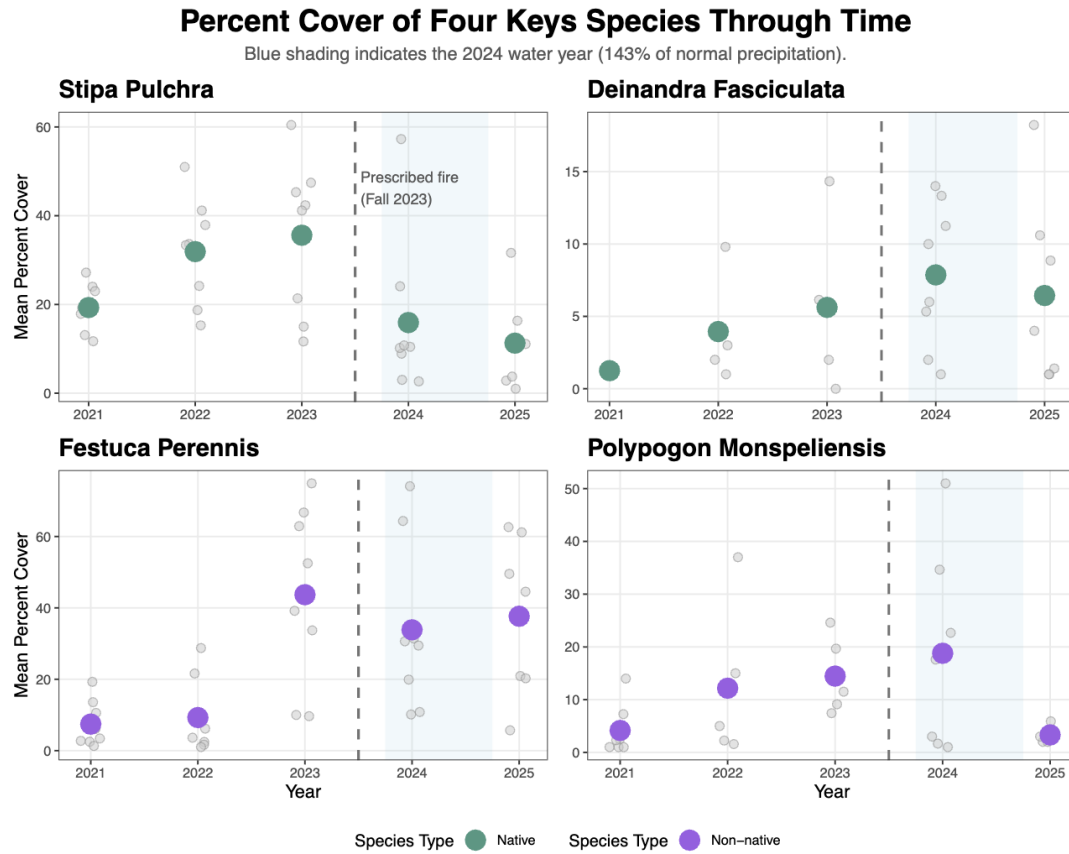


Figure 6: Figure 6

Discussion

References

Rickard, A. 2024. "North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024." *UC Santa Barbara: Cheadle Center for Biodiversity and Ecological Restoration*. <https://escholarship.org/uc/item/16p046mb>.